Benjamin Kim
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Google Scholar
Website

Education

Massachusetts Institute of Technology — Cambridge, MA
Ph.D. in Electrical Engineering and Computer Science — *Jun. 2026 – Present*

University of Illinois Urbana-Champaign — Urbana, IL
M.S. in Electrical and Computer Engineering — *Aug. 2025 – May 2026*

University of Illinois Urbana-Chamapaign — Urbana, IL
B.S. in Computer Engineering, Minor in Mathematics — *Aug. 2021 – May 2025*

Research Experience

Massachusetts Institute of Technology — Cambridge, MA
Graduate Research Assistant | Supervised by Muriel Médard — *Jun. 2026 – Present*

Harvard University — Cambridge, MA
Research Fellow | Supervised by Flavio du Pin Calmon — *Jun. 2026 – Present*

Amazon Science — Remote
Student Collaborating Researcher — *Jan. 2026 – Present*

UIUC Coordinated Science Laboratory — Urbana, IL
Visiting Graduate Researcher | Supervised by Daniel Alabi — *May. 2026 – Present*
Graduate Research Assistant | Supervised by Lav Varshney and Daniel Alabi — *Aug. 2025 – May 2026*

MIT Lincoln Laboratory — Lexington, MA
Research Intern | Tactical Edge Communications Group 62 — *Jun. 2025 – Sep. 2025*

MIT Research Laboratory of Electronics — Cambridge, MA
Undergraduate Research Assistant | Supervised by Muriel Médard — *Jun. 2022 – Aug. 2025*

Awards and Fellowships

Irwin and Joan Jacobs Presidential Fellowship, MIT — Mar. 2026
NSF Graduate Research Fellowship, National Science Foundation — Apr. 2025
Edward C. Jordan Award, for outstanding undergraduate academics and research — Apr. 2025
CRA Outstanding Undergraduate Researcher Award Finalist 1 of 30 in NA — Jan. 2025
UIUC Outstanding Senior (Homecoming Court) Finalist 40 out of 9000 seniors — Jul. 2024
Donald L. Bitzer and H. Gene Slottow Creativity Award — Apr. 2024

Publications

[11] CARDS: Coded Aggregation for Robust and Differentially Private Shuffling
Benjamin D. Kim, Muriel Médard, Daniel Alabi
Submitted to IEEE Journal on Selected Areas in Information Theory, 2026

[10] SimpleMark: a Fast Multi-Bit Text Watermark with Tunable f-Divergence Security
Benjamin D. Kim, Wanrong Zhang, Weitong Ruan, Lav R. Varshney, Daniel Alabi
Submitted to ICLR, 2027

[9] Optimal Guarantees for Auditing Rényi Differentially Private Machine Learning
Benjamin D. Kim, Lav R. Varshney, Daniel Alabi
Submitted to ICLR, 2027
TPDP 2026 and ICML Workshop on Hypothesis Testing

[8] Scalable Multiterminal Key Agreement via Error-Correcting Codes
Benjamin D. Kim, Daniel Alabi, Lav R. Varshney
IEEE Information Theory Workshop (ITW), 2026

[7] Revisiting the Interface between Error and Erasure Correction in Wireless Standards
Vipindev Adat Vasudevan, Homa Esfahanizadeh, **Benjamin D. Kim**, Laura Landon, Alejandro Cohen, Muriel Médard
IEEE Journal on Selected Areas in Communications, 2026

[6] An Efficient Hybrid Key Exchange Mechanism
Benjamin D. Kim, Vipindev Adat Vasudevan, Alejandro Cohen, Rafael G. L. D'Oliveira, Thomas Stahlbuhk, Muriel Médard
IEEE Information Theory Workshop (ITW), 2025

[5] On the Benefits of Coding for Network Slicing
Homa Esfahanizadeh, Vipindev Adat Vasudevan, **Benjamin D. Kim**, Shruti Siva, Jennifer Kim, Alejandro Cohen, Muriel Médard
IEEE ICC Workshops, 2024

[4] CRYPTO-MINE: Cryptanalysis via Mutual Information Neural Estimation
Benjamin D. Kim, Vipindev Adat Vasudevan, Jongchan Woo, Alejandro Cohen, Rafael G. L. D'Oliveira, Thomas Stahlbuhk, Muriel Médard
IEEE ICASSP, 2024

[3] Cryptanalysis via Machine Learning Based Information-Theoretic Metrics
Benjamin D. Kim, Vipindev Adat Vasudevan, Rafael G. L. D'Oliveira, Alejandro Cohen, Thomas Stahlbuhk, Muriel Médard
Preprint, 2025

[2] Leveraging AES Padding: dBs for Nothing and FEC for Free in IoT Systems
Jongchan Woo, Vipindev Adat Vasudevan, **Benjamin D. Kim**, Rafael G. L. D'Oliveira, Alejandro Cohen, Thomas Stahlbuhk, Ken R. Duffy, Muriel Médard
Preprint, 2024

[1] CERMET: Coding for Energy Reduction with Multiple Encryption Techniques — It's easy being green
Jongchan Woo, Vipindev Adat Vasudevan, **Benjamin D. Kim**, Rafael G. L. D'Oliveira, Alejandro Cohen, Thomas Stahlbuhk, Muriel Médard
Preprint, 2023

Talks and Posters

Cryptanalysis via Mutual Information Neural Estimation

- Talk at University of Illinois Electrical and Computer Engineering Senior Thesis Symposium — Dec. 2024
- Poster at IEEE International Conference on Acoustics, Speech and Signal Processing 2024 — Apr. 2024
- Poster at 2024 UIUC Undergraduate Research Symposium — Apr. 2024
- Invited talk at 2024 Information Theory and Applications Workshop (ITA) — Feb. 2024

Efficient Hybrid Key Exchange Mechanisms

- Talk at IEEE Information Theory Workshop 2025 — Sep. 2025

Auditing Rényi Differentially Private Machine Learning

- Invited poster at 2026 Information Theory and Applications Workshop (ITA) — Feb. 2026
- Invited poster at Coordinated Science Laboratory Annual Conference — Feb. 2026
- Invited poster at CSL 75th Anniversary Research Showcase — Mar. 2026
- Accepted poster at TPDP Workshop 2026 — Jun. 2026
- Accepted poster at ICML Workshop on Hypothesis Testing — Jul. 2026

Teaching and Outreach

Teaching Assistant for Algorithms & Models of Computation — Dec. 2023 – Present

- CS/ECE 374, Algorithms course at UIUC
- Graduate Teaching Assistant (Fall 2025, Spring 2026)
- Undergraduate Teaching Assistant (Spring 2024, Fall 2024, Spring 2025)

Allerton Conference Session Chair — Sep. 2025

- Ran Network Engineering session and intorduced speakers